

Build and Diagnose a Custom VPC

90 minutes · Mapped criteria A3, K4

Purpose and work product

Build a custom VPC and regional subnets, apply narrow firewall rules and diagnose a failed connection.

Submit: network inventory, connectivity-test result and firewall evidence

Safety, cost and privacy

Use only the assigned training project. Confirm the active account, project and region before every mutation. Never place credentials, payment details, access tokens or private keys in screenshots or submitted files. Stop billable resources and verify the final inventory.

Files in this folder

File	Purpose
network_setup.sh	Mock data, script or configuration used by this activity
network_plan.csv	Mock data, script or configuration used by this activity

Detailed procedure

Step 1

Set the correct project and region asia-southeast1.

Step 2

Review network_plan.csv and confirm neither CIDR overlaps the default network or an assigned corporate range.

Step 3

Run network_setup.sh to create ace-vpc and app-sg.

Step 4

Create data-sg from the plan using a separate gcloud command.

Step 5

List the VPC, subnets, routes and firewall rules; save the tables as network-inventory.txt.

Step 6

Create two small test VMs without external IP addresses, one tagged web and one untagged.

Step 7

Serve a test page on the tagged VM and test TCP 80 from the second VM.

Step 8

Remove the web tag temporarily and repeat the test. Record the failure and identify the target mismatch.

Step 9

Restore the tag and create a Network Intelligence Center Connectivity Test between the endpoints.

Step 10

Enable or inspect firewall logging and preserve the allow/deny evidence.

Step 11

Stop the VMs after testing; keep the network for later activities.

Verification record

Record expected and observed output. A successful command without the correct project, identity, scope or evidence is not an accepted result.

Check	Expected evidence
Identity and scope	Active account, project ID, region or zone
Configuration	Resource description or policy output
Runtime result	Test output with UTC timestamp
Failure test	One controlled negative test and diagnosis
Cleanup	Final inventory and stopped/deleted billable resources

Troubleshooting sequence

Check the active identity and project first, then API enablement and IAM, then resource state, networking and quotas. Preserve the failed output before making a correction. If the issue remains, escalate with the exact command, timestamp, expected result, observed result and evidence files.

Cleanup

Delete or stop resources as instructed. Re-open Billing and the relevant service inventory, then record the remaining resource names. Budget alerts do not stop charges.